

Al (Al I). Strongest classified transitions below 2 μm (NIST): 33 levels, 45 lines.

Al⁺ ionisation limit 48278.48 cm⁻¹ = 5.98577 eV (207.132 nm)



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \text{er} J' \rangle $ (ea0)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$3p\ ^2P_{3/2}^o - 3d\ ^2D_{5/2}$	309.3606	309.2708	969.0712	2.495	7.100e+07	11.37	99.4	309.4 nm laser-cooling transition $3p\ 2P_{3/2}$
$3p\ ^2P_{1/2}^o - 4s\ ^2S_{1/2}$	394.5122	394.4006	759.9066	1.688	4.700e+07	23.23	32.2	394.4 nm resonance line
$3p\ ^2P_{3/2}^o - 4s\ ^2S_{1/2}$	396.2641	396.1520	756.5471	2.466	9.900e+07	23.23	67.8	396.2 nm line

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

²⁷Al hyperfine structure ($I = 5/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
$3p\ ^2P_{3/2}^o$	502.0346(5)	-	F=2: -878.56	F=3: +627.54		
$3p\ ^2P_{3/2}^o$	94.27723(10)	18.915	F=1: -481.71	F=2: -308.29	F=3: -33.97	F=4: +358.27
$4s\ ^2S_{1/2}$	431.11(9)	-	F=2: -754.44	F=3: +538.89		
$4p\ ^2P_{3/2}^o$	58.280	-	F=2: -101.99	F=3: +72.85		
$4p\ ^2P_{3/2}^o$	23.120	2.860	F=1: -119.38	F=2: -75.43	F=3: -7.35	F=4: +87.42

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$3p\ ^2P_{3/2}^o - 4s\ ^2S_{1/2}$	396.264	26m-27	378(3)
$3p\ ^2P_{3/2}^o - 4s\ ^2S_{1/2}$	396.264	28m-27	-321.9(8)
$3p\ ^2P_{3/2}^o - 4s\ ^2S_{1/2}$	396.264	29-27	-610.7(6)
$3p\ ^2P_{3/2}^o - 4s\ ^2S_{1/2}$	396.264	30-27	-889.5(12)
$3p\ ^2P_{3/2}^o - 4s\ ^2S_{1/2}$	396.264	31-27	-1141.0(12)
$3p\ ^2P_{3/2}^o - 4s\ ^2S_{1/2}$	396.264	32-27	-1401(7)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope})$$